MOLECULAR DYNAMICS EVALUATION OF MECHANISTIC PROPERTIES DRIVING PASSIVE PERMEATION IN BINARY BILAYERS

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ABSTRACT

While passive permeation is important for pharmacokinetics, the thermodynamic impact of biological membrane complexity remains poorly mapped. We systematically evaluated how lipid bilayer composition shapes the free energy profiles of six diverse drug-like xenobiotics (amantadine, diclofenac, metoprolol, pindolol, ranitidine, scopolamine). Using the MemCross all-atom molecular dynamics (MD) protocol[1] and Accelerated Weight Histogram (AWH) method, we simulated permeation across five binary lipid mixtures representative of human plasma membranes.

Our findings reveal that permeation thermodynamics are co-driven by specific lipid structural features and the intrinsic molecular properties of the permeants. We developed multivariate models capable of predicting key thermodynamic parameters based on mechanistic descriptors quickly obtainable from MD simulations.

REFERENCES

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